Project 2: Student Management System (CRUD)

In this project, we will develop a simple **Student Management System (SMS)** using Python. This system will allow the user to perform CRUD operations—Create, Read, Update, and Delete—on student records. Each student will have a name, roll number, and other essential information such as age and grade. The program will interact with the user through the console and use a list or dictionary to store student records. We will also implement functions to add, view, update, and delete student details from the system.

This project will give you hands-on experience with fundamental concepts such as:

- **CRUD operations**: Creating, reading, updating, and deleting data.
- **Data management**: Using Python data structures (e.g., lists or dictionaries).
- **User input handling**: Receiving and processing input from users.
- **Control flow**: Using loops and conditionals to control the flow of the program.
- **Error handling**: Managing invalid inputs and exceptions.

Step-by-Step Guide to Build the Student Management System (CRUD)

Step 1: Define the Data Structure

The student records can be stored in a dictionary, where each key is a unique roll number and each value is another dictionary holding the student's information. This will allow us to efficiently manage, update, and delete student records.